

GUSTAVO FLORINDO OLIVEIRA

APPLIED COMPUTER VISION ENGINEER | VIDEO INTELLIGENCE & ML SYSTEMS

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PROFESSIONAL SUMMARY

Applied Computer Vision Engineer with 5+ years of software engineering experience and hands-on delivery of production video-intelligence systems. Built Python/YOLO solutions for Petrobras maritime safety, including fallen-person detection, PPE compliance monitoring, and danger-zone violation detection from vessel camera footage. Experienced in improving computer-vision behavior in multi-person scenes, diagnosing failure modes, integrating GPU-accelerated video pipelines, and supporting containerized ML services in production. Strong foundations in Python, PyTorch, OpenCV, FFmpeg, backend engineering, distributed messaging, and cloud infrastructure.

COMPUTER VISION HIGHLIGHTS

- **Production video analytics:** delivered real-world safety monitoring workflows for maritime camera streams in the Duvens Analytics platform for Petrobras.
- **Challenging visual conditions:** worked with changing viewpoints, lighting, occlusion, environmental variation, and crowded multi-person scenes.
- **Model improvement:** investigated computer-vision failure cases, fixed defects, and improved accuracy while preserving system reliability through tests and validation.
- **Production integration:** supported H.264/FFmpeg video processing, GPU inference, containers, messaging pipelines, monitoring, and operational troubleshooting.

TECHNICAL SKILLS

Computer Vision & ML: YOLO, PyTorch, TensorFlow, OpenCV, Roboflow, object detection, video analytics, multi-person scene analysis, event/safety detection, model evaluation, failure-mode analysis, transfer learning, OCR, image preprocessing

Video & GPU Workflows: FFmpeg, H.264, GPU-accelerated inference, video-stream processing, production inference pipelines

Geometric Computer Vision: Working knowledge of camera calibration, homography, projective geometry, coordinate transformations, and image-to-world mapping

ML / Software Engineering: Python, NumPy, Pandas, FastAPI, Django, REST APIs, unit testing, regression testing, maintainable production code

Infrastructure & Data: Docker, Kubernetes, RabbitMQ, Kafka, AWS, GCP, Terraform, CI/CD, Databricks, SQL, PostgreSQL, MySQL, Oracle, MongoDB

PROFESSIONAL EXPERIENCE

Backend Developer - Computer Vision Systems - TELNAV

Feb 2026 - May 2026

Remote, Contract

- Developed computer-vision solutions for Duvens Analytics, a maritime safety analytics platform delivered for Petrobras, using Python, YOLO, and Roboflow.
- Built detection workflows for fallen individuals, PPE compliance, and danger-zone violations in real-world vessel camera footage.
- Integrated H.264 video processing and FFmpeg-based workflows with GPU-accelerated inference for operational safety monitoring.
- Adapted detection logic for variable camera angles, lighting, occlusion, environmental conditions, and multiple people in the scene.
- Translated safety rules into maintainable application logic and collaborated with engineering stakeholders to prepare solutions for production use.

Backend Developer - Computer Vision Systems - Noleak Defence

May 2024 - Present

Remote

- Investigated failure cases and improved computer-vision accuracy in multi-person scenarios, with emphasis on practical behavior in production conditions.
- Fixed application and vision-pipeline defects while adding unit tests and regression safeguards to preserve system integrity during changes.
- Monitored containerized services and messaging infrastructure supporting production workloads and operational support.
- Built message producers and validation components for messaging pipelines, improving throughput and processing efficiency.
- Worked independently across debugging, testing, performance optimization, deployment support, and production troubleshooting.

SELECTED COMPUTER VISION & IMAGE PROCESSING WORK

- Video and image restoration: analyzed Fourier spectra and implemented Gaussian notch filtering to attenuate periodic noise while balancing artifact removal and detail preservation in MATLAB.
- Interactive object segmentation: built HSV-based workflows to separate objects from the background and classify a user-selected color under lighting variation, including circular hue-distance handling.
- Depth-image scene segmentation: applied median filtering, intensity/spatial masks, connected-component analysis, and pseudo-color visualization to isolate people and structural regions.
- Gesture-recognition research: studied computer-vision approaches for Brazilian Sign Language (Libras) in educational robotics, including gesture recognition, hand/pose analysis, dataset limitations, and real-world deployment challenges.

ADDITIONAL SOFTWARE ENGINEERING EXPERIENCE

Full Stack Developer - Social Place Tecnologia Jan 2024 - Feb 2025

Brazil

- Developed backend APIs and analytical CRM features for targeted advertising, including pagination and search optimization.
- Built data-driven functionality to support operational and business decision-making.

Full Stack Developer - Zigurate Feb 2022 - Nov 2024

Remote

- Built and maintained management platforms, automated tests, and asynchronous workflows using RabbitMQ.
- Contributed to modular, maintainable frontend and backend systems in a remote product environment.

Analyst / Frontend Developer - Vortex Labs Solutions Feb 2024 - Mar 2025

Remote

- Defined project scopes, translated client requirements into technical flows, documented APIs, and built React/Redux interfaces.

Database Administrator - Conexos Sep 2021 - Dec 2022

Brazil

- Optimized database load times by 65%, developed reporting queries, and supported large-scale data migrations.

Python Web Scraper - Scoutify Feb 2020 - Nov 2020

United States, Remote

- Automated data collection from sports-industry websites using Python and organized datasets with Pandas and Databricks.
- Built repeatable data workflows, providing early experience with sports-domain data and international remote collaboration.

EDUCATION

Systems Analysis and Development - Federal Institute of Esp rito Santo 2020 - 2023

Brazil

LANGUAGES

English: Advanced (C1) | Portuguese: Native / Fluent